

Summary

PhD researcher and engineer building e3nn/PyTorch Geometric equivariant models and JAX/PyTorch computational graphs for materials simulation. Experience spans automatic differentiation, GPU acceleration, distributed sharding, and national-scale HPC systems.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed e3nn and PyTorch Geometric equivariant graph neural networks, DeePH-style computational graphs, and automatic-differentiation models to accelerate material-property and optical calculations.
- Built distributed and accelerator-enabled ML software in Python and C++ using JAX/PyTorch computational graphs, CUDA, ROCm/HIP, MPI, and OpenMP; sharded workloads across up to 800 Frontier GPUs and TB-scale datasets.
- Applied machine learning models alongside first-principles materials simulations on supercomputing clusters, connecting model development with production scientific-computing workloads.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding ML work in real materials-science applications.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built GPU-accelerated processing and visualization tools for large waveform datasets, providing practical experience with high-throughput data pipelines and performance-oriented engineering.
- Built engineering-support data-management tooling with C# and ASP.NET to collect, organize, and process field-tool datasets.

Selected Projects

Materials ML for Optical Properties: Developed e3nn/PyTorch Geometric equivariant graph neural networks and automatic-differentiation models for material energies and optical properties; integrated the models with 800-GPU, TB-scale workflows.

Agentic Scientific RAG: Built retrieval-driven tooling that reads documentation and research papers to generate structured inputs for computational science software.

Container Orchestration for First-Principles Computing: Developed CUDA-ready and cross-environment containers for first-principles software stacks and published them to Docker Hub for portable scientific computing.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, transformers, automatic differentiation

Programming: Python, C++, CUDA, HIP, JAX, PyTorch, Bash

Data / Compute: NumPy, pandas, SciPy, JAX/PyTorch graphs, distributed sharding, CUDA/ROCm/HIP, TB-scale data

Systems: Linux, Docker, Docker Compose, Kubernetes, gcloud

Scientific: PETSc, SLEPc, first-principles simulation, materials modeling, equivariant graph neural networks, physics-informed ML